

State of the Map Europe 2025



MapSwipe Today and its Future in OSM

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Who here has...

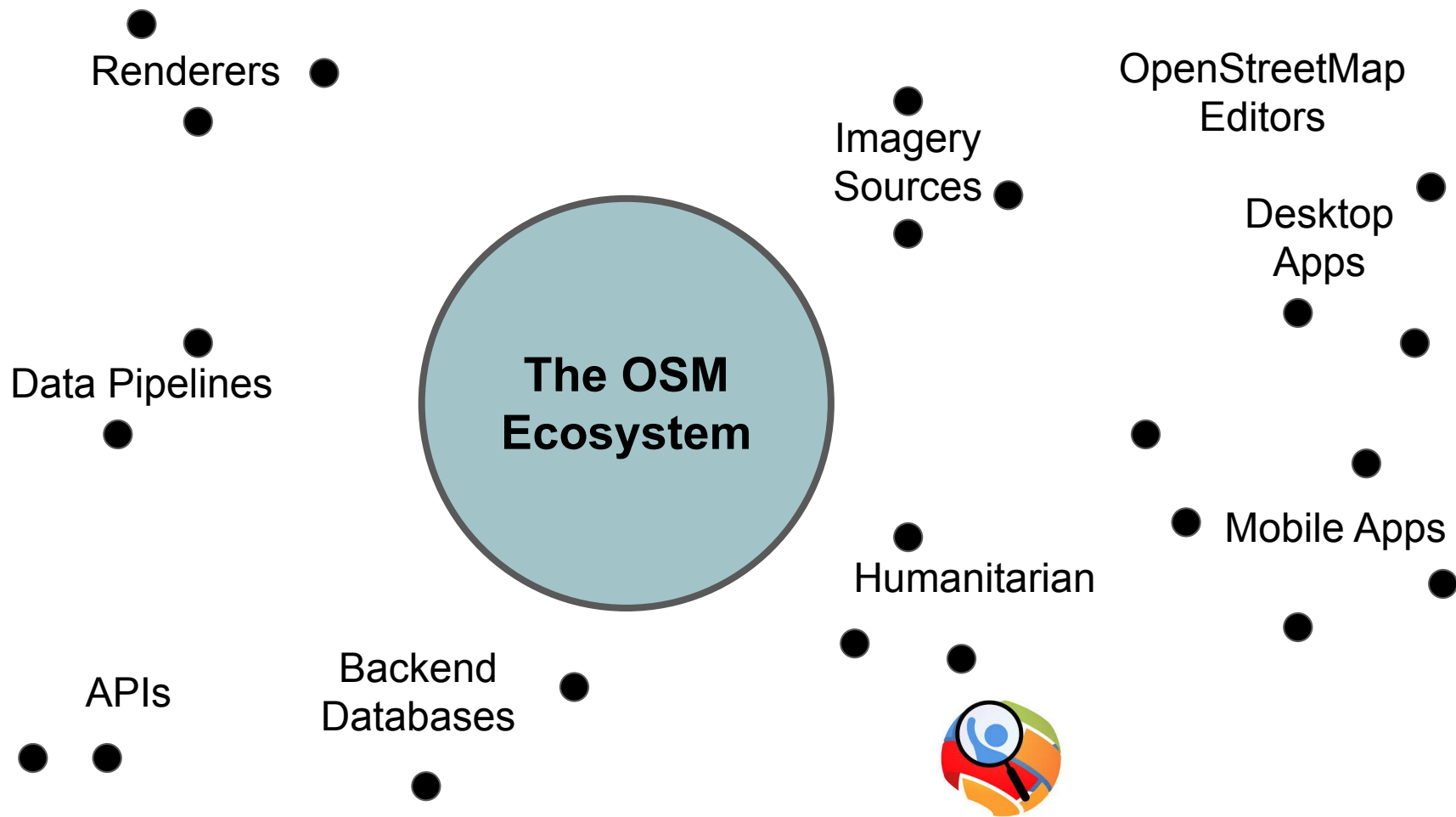
- Heard of MapSwipe (but never used it)
- Has used MapSwipe to map
- Has created a mapping project on MapSwipe
- Has never heard of MapSwipe

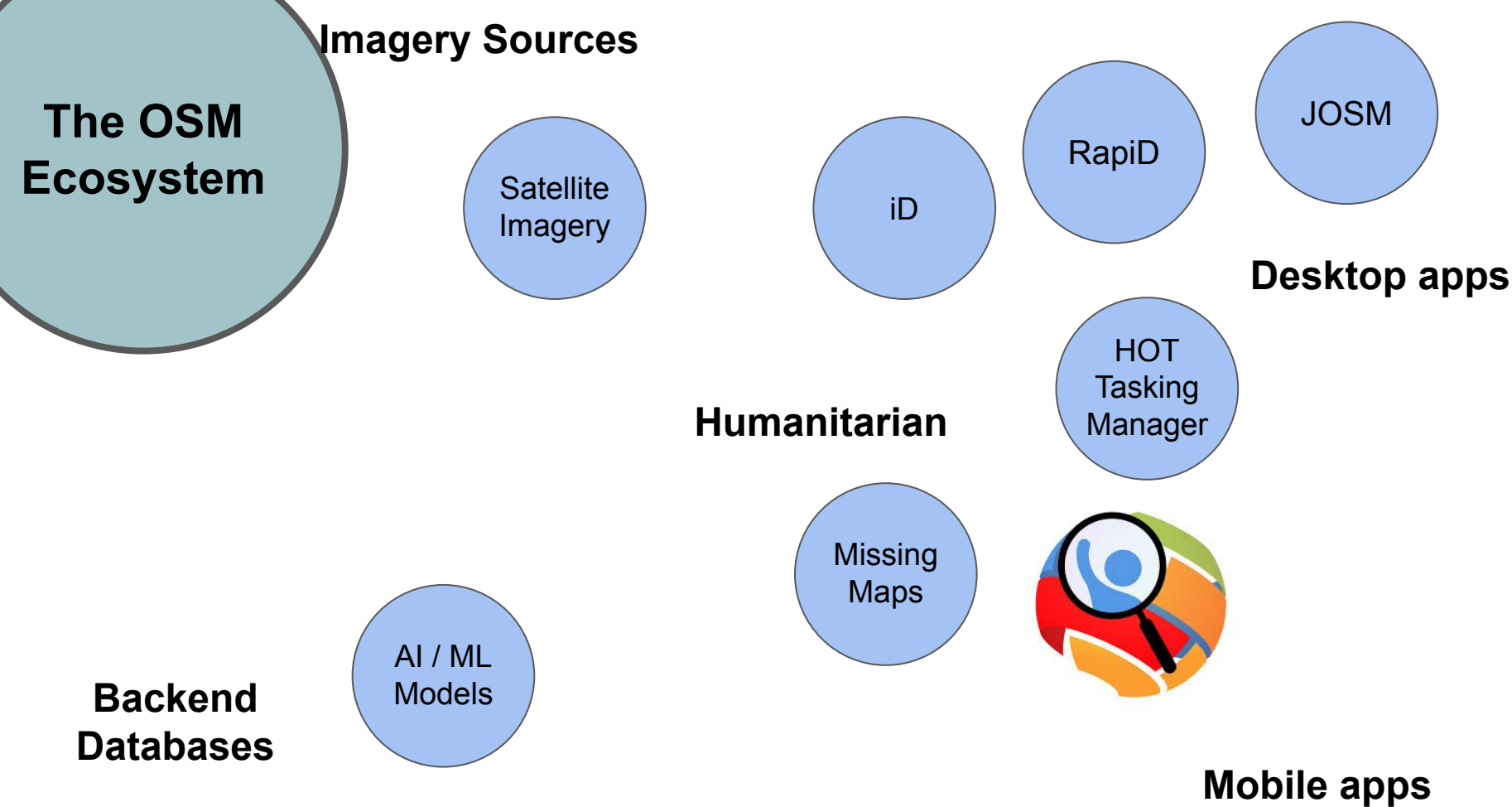
What is MapSwipe?

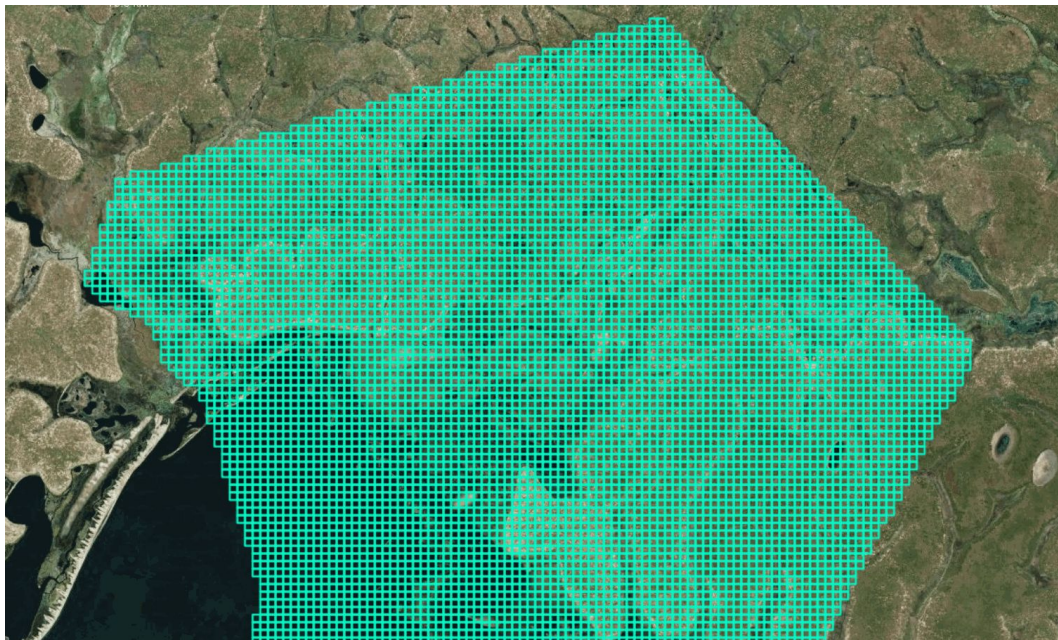


MapSwipe









MapSwipe project for locating buildings in rural Zambia. MapSwipe can reduce the area for detailed mapping in remote places **by over 99%**



**Missing
Maps**



MapSwipe

Free open source web & mobile application launched by MSF in 2016, under the Missing Maps umbrella.

Volunteer-driven and community-led project to support Missing Maps. Core partner organizations also include HOT, HeiGIT, ARC, BRC

MapSwipe has an elected Governance team to make big decisions, particularly on spending, prioritization, creating roadmaps for future development



State of the Map Europe
14th & 15th November 2025 Dundee, Scotland, UK

Important Bits

- MapSwipe is not currently an OSM editor - it has its own database!
- At least three users look at each tile / task
- MapSwipe is both a mobile phone app and web app (use it in the browser)

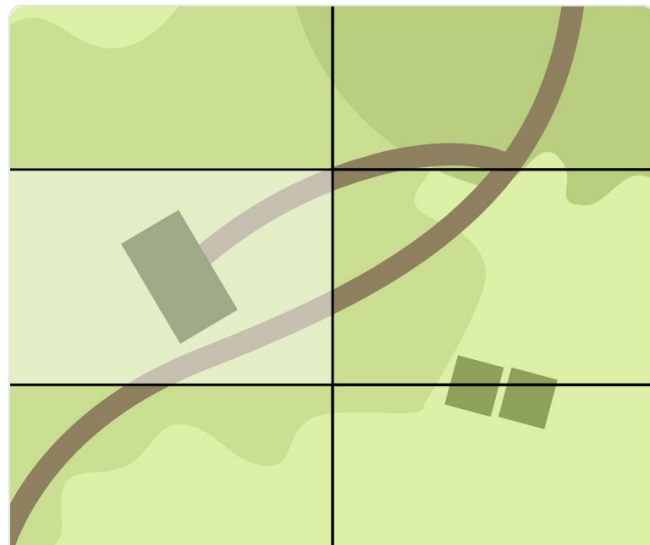
Current Day MapSwipe

Project Types



The Find Features Project Type

- The classic MapSwipe project type
- Users search for a feature
 - Buildings
 - Roads
 - Airline strips
 - Mobile homes
 - Mangroves
 - Solid waste
 - Anything you can see from a satellite!



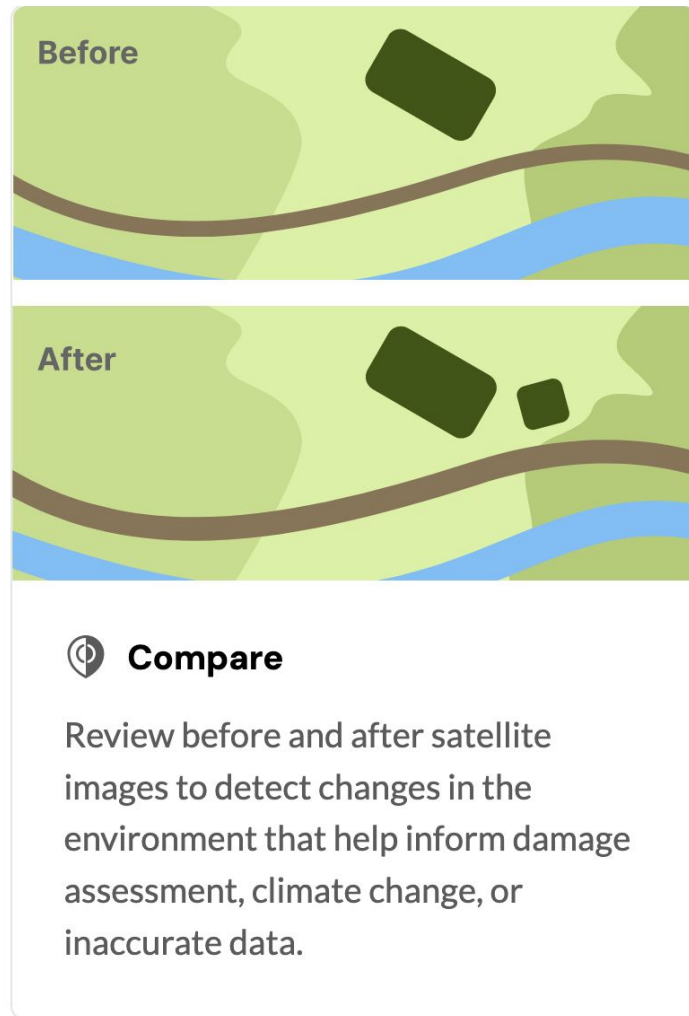
Find

Swipe through satellite images to identify & select those that contain the requested features such as buildings, roadways, waterways and more.



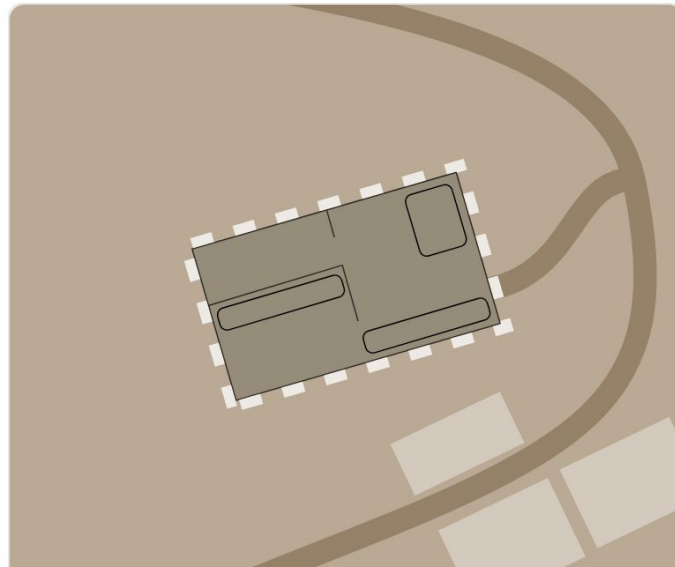
The Compare Dates Project Type

- Users compare before and after scenes to indicate a change
- Can be used for before / after disasters
- Can be used to see how cities have built up (or diminished) over time
- Least common MapSwipe project type (difficult to find imagery)



The Validate Footprints Project Type

- Users look at existing OSM data
- Can pull directly from projects on the HOT Tasking Manager
- Can help humanitarian organizations to quickly see which areas need to be remapped



Validate

Assess building footprints for accuracy where buildings have been previously traced by remote mappers or through AI to identify where remapping is needed.



The View Streets Project Type (web only) (experimental)

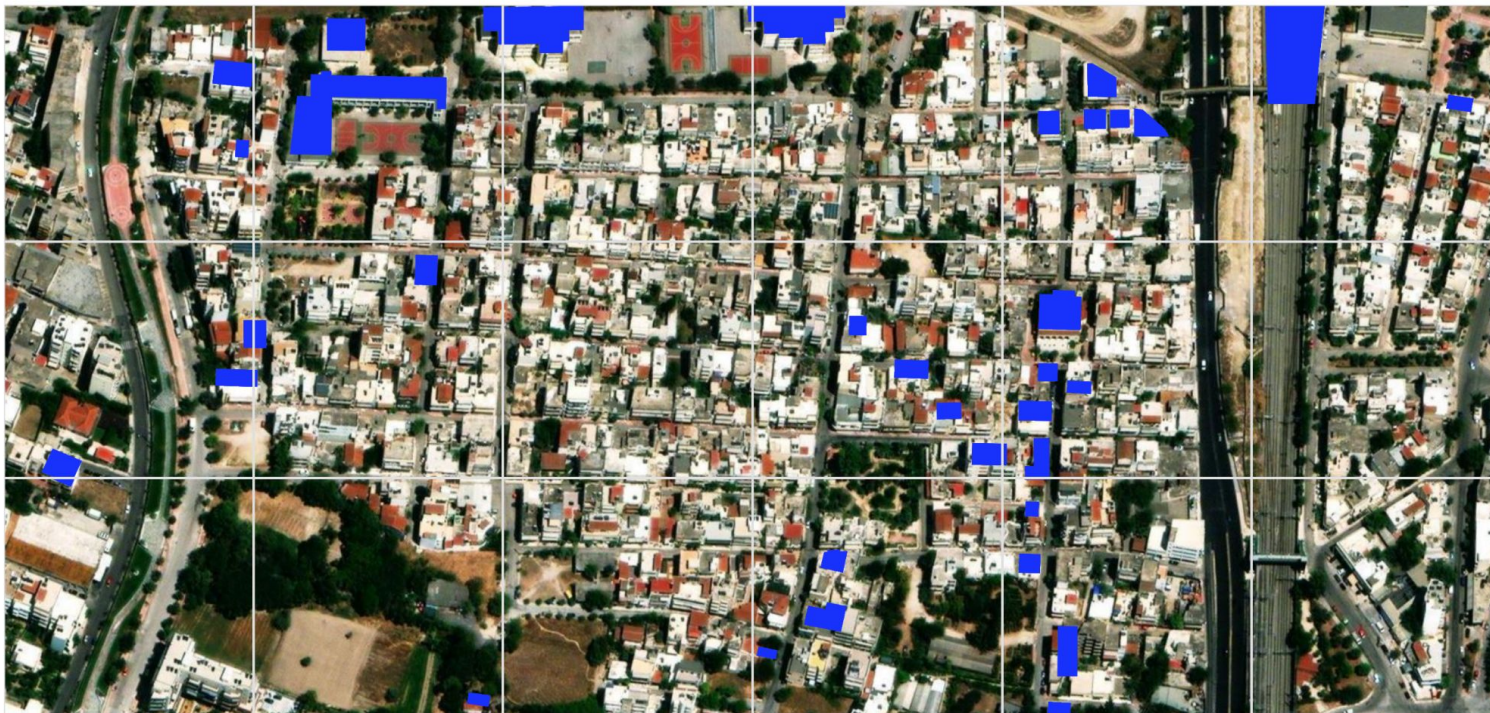
- Users look at street-level imagery to look for aspects of a feature
 - Is the road paved?
 - Is solid waste in the image?
 - Is this building a school?
 - What is the surface quality of the road?

Waste Mapping

You are looking for: waste



The Check Completeness Project Type



Compare the buildings layer with the image tiles.

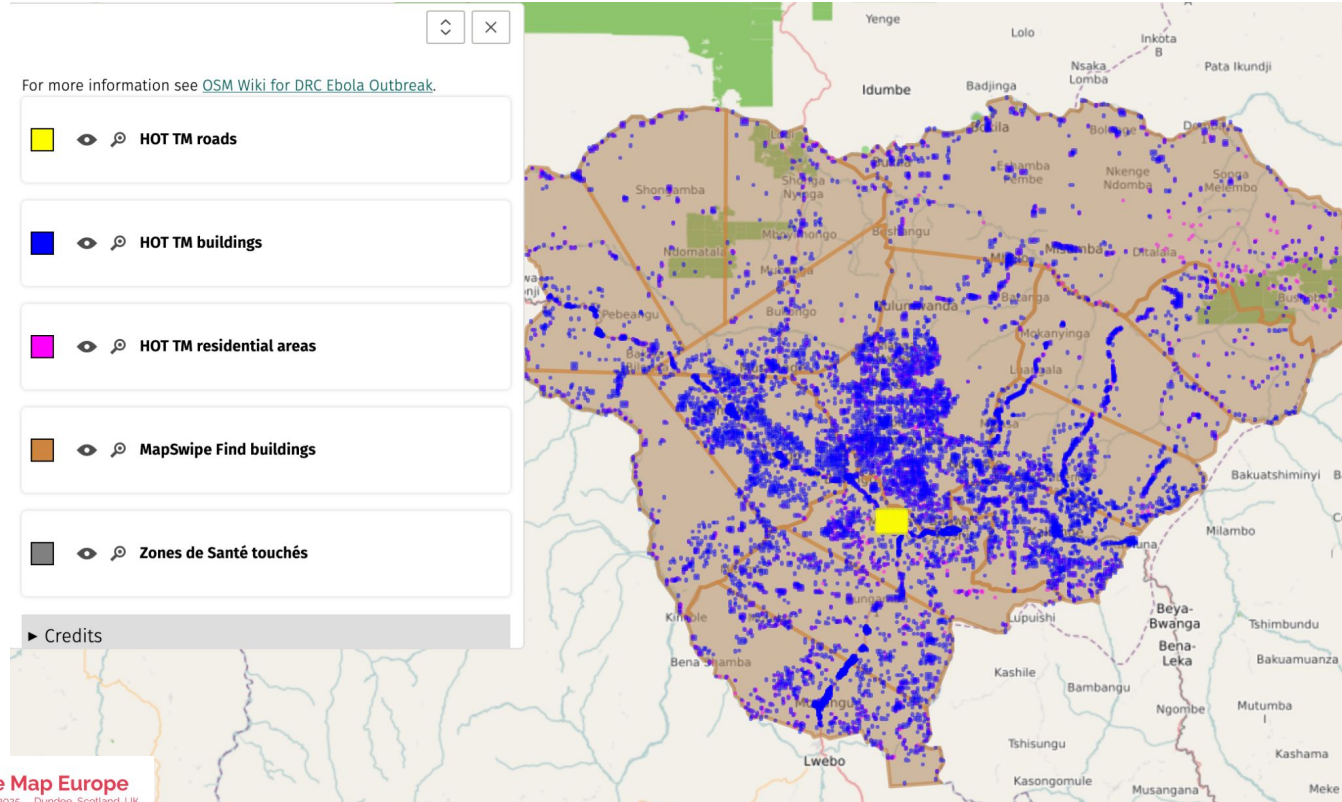


Answering questions about **any** image collection, not just tiles from a map service of aerial imagery. Optionally, with data about object detections & inferences to validate.

Real Life Recent MapSwipe Use Cases



2025 - Democratic Republic of Congo Ebola Outbreak (MSF)



2025 - Democratic Republic of Congo Ebola Outbreak (MSF)



2025 - Myanmar Community-led Damaged Buildings Mapping (May - Oct)

#23733 | Open Mapping Hub Asia-Pacific

LOW

BUILDING DAMAGE ASSESSMENT IN MAHAAUNGMYAY, MYANMAR [REMOTE]

ARCHIVED

Myanmar Earthquake 2025 · Myanmar

INSTRUCTIONS

⚠ This project is archived and read-only. You can view the project, but you cannot update tasks.

Imagery Selection

1. Primary Option: : Use **default Imagery** from Maxar as the default choice for clarity and updates.
2. **Activate GPS Traces:**



2025 - Myanmar Community-led Damaged Buildings Mapping

Use **before and after imagery** to compare. Tag the building **only if**:

- It was **not damaged before**, and **damaged after** the disaster.

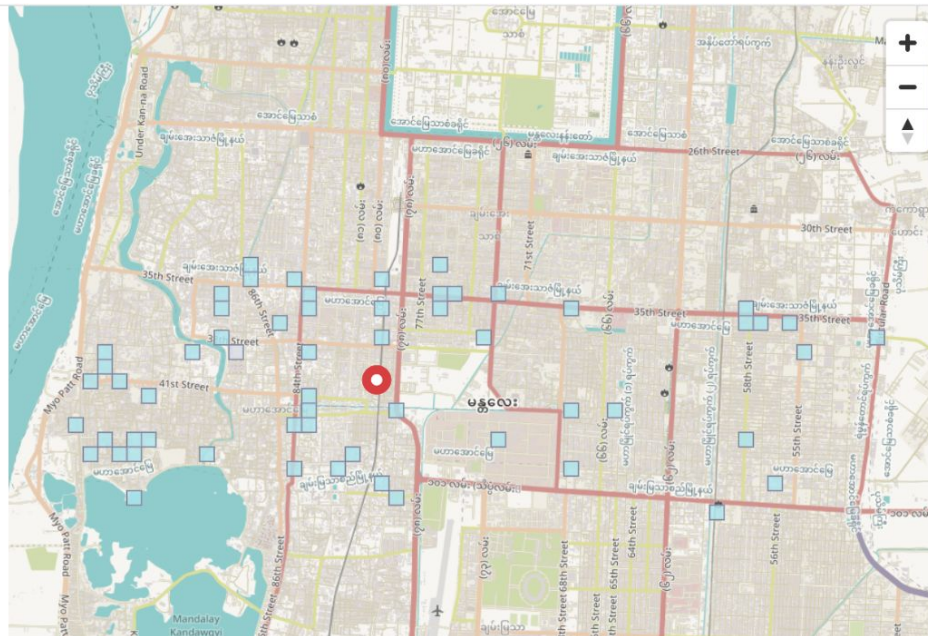
🚫 **Do not tag if:**

- It was damaged both before and after.
- It was not damaged before or after.
- It was damaged before but appears repaired or undamaged after.

The objective is to avoid mapping prior damages and erroneously attributing the damage to the earthquakes.

📌 What Tags to Use

- `damage=yes`
- `damage:event=2025MyanmarEQ`
- `source:damage:date=2025-03-31`
- `source:damage=Maxar`



MapSwipe's Structure





open collective

MapSwipe Working Group

Every second Tuesday

USA - PT	USA - MT	USA - CT	USA - ET	UK	Germany
8am	9am	10am	11am	4pm	5pm

Community
+ tool

What's next for MapSwipe

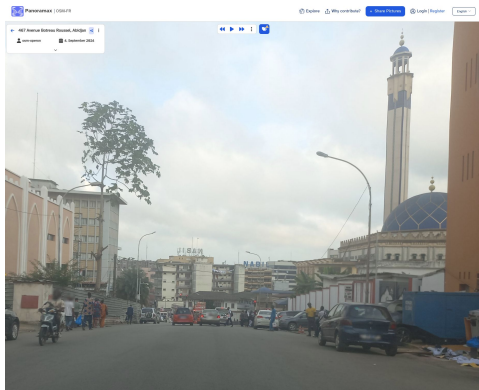


MapSwipe View Streets x Panoramax - coming soon!

Panoramax as additional image provider for View Streets projects

Panoramax:

- Street-level image resource similar to Mapillary
- Fully open source, self-hosting possible
- Active community development
- Federation of instances
- Meta-catalog at api.panoramax.xyz



With MapSwipe:

Classify Panoramax images in a specific area and time range.

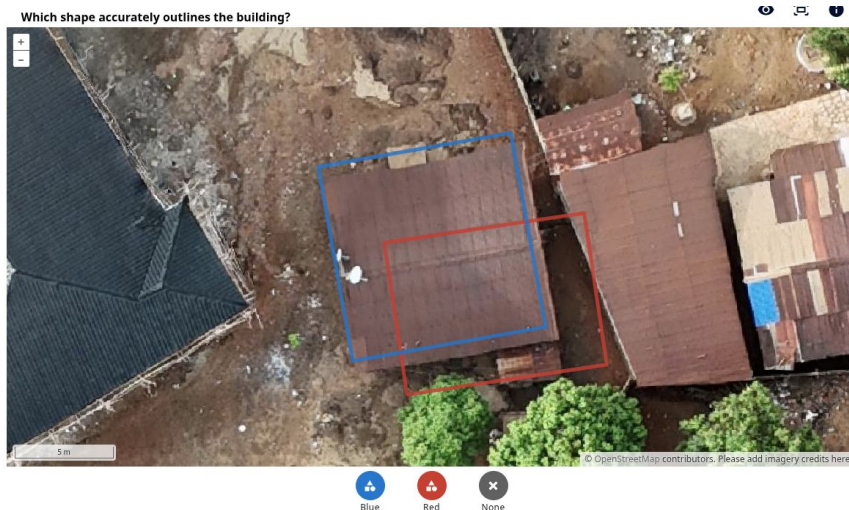
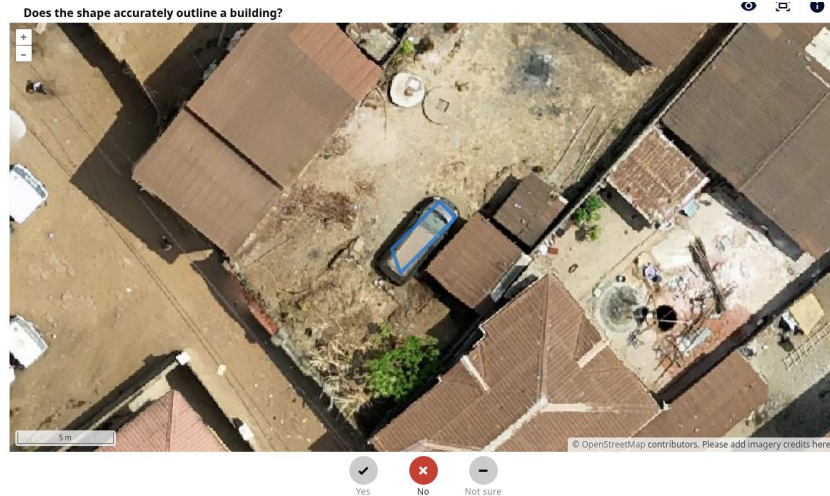
Crowd-sourced data can be useful in itself, or can be used for pre-selection of images or as labels in Machine Learning workflows.

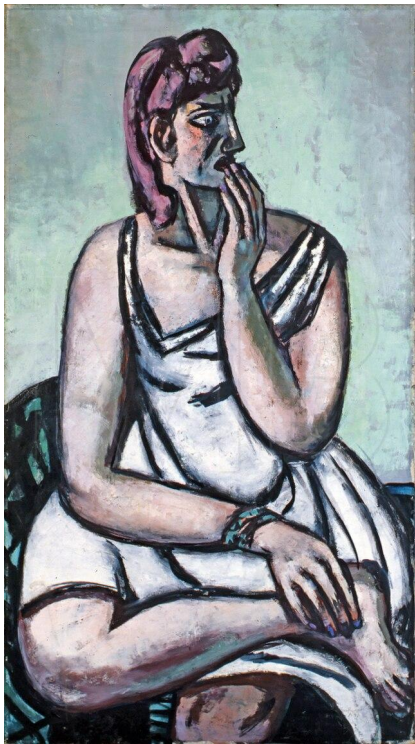
New project type: Conflation

Compare building footprints from OSM and another dataset

Volunteers choose between:

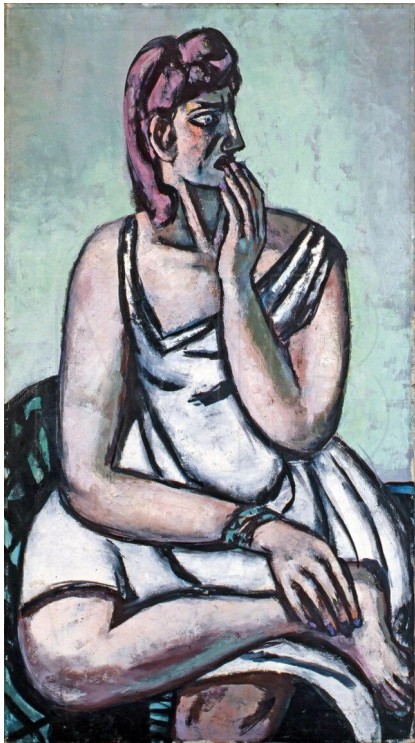
- If only **one** building outline exists, validate if it is accurate against the shape of the building in the imagery (yes, no, not sure)
- If **two** building outlines exist, validate outline better fits the shape of the building (red, blue, none)
- Volunteer will **not know which** is OSM data and which is another dataset





Big Questions





Q1. Should MapSwipe
become its own OSM
editor?



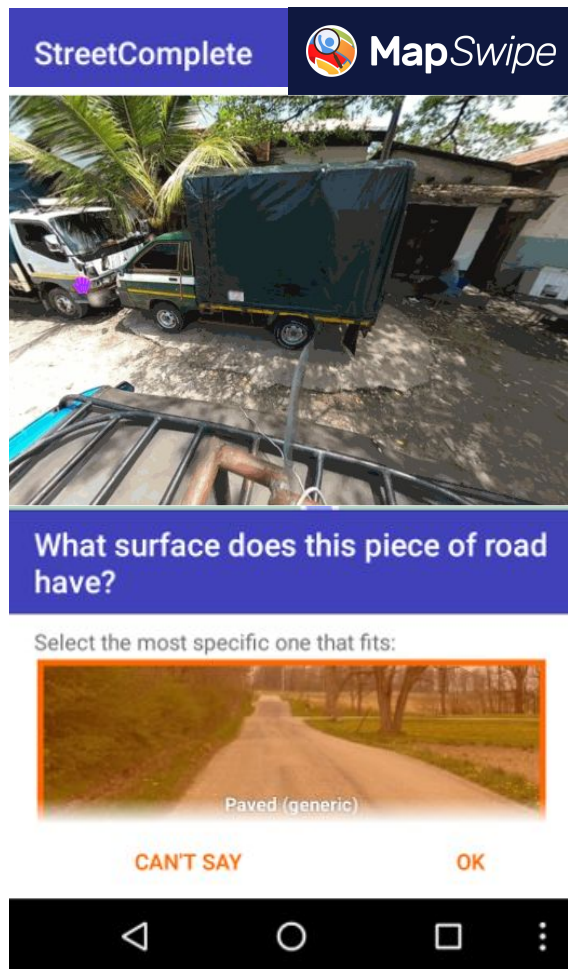
Human validated attributes added to existing OSM features

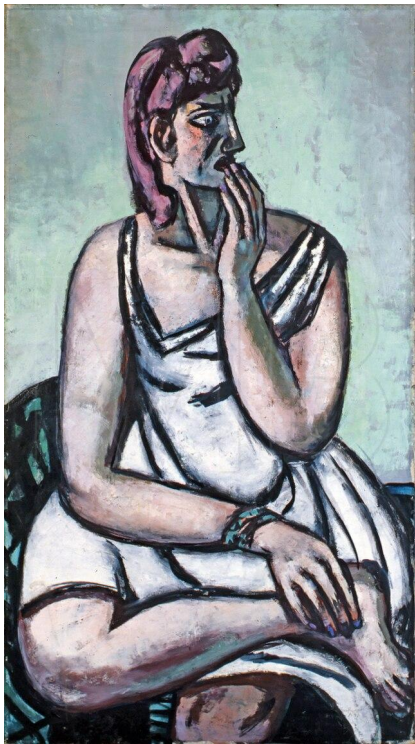
“If all 3+ people say the same thing on MapSwipe, isn’t it pretty accurate?”

“Like the StreetComplete app, but lets you see the street level imagery remotely.”

So kinda like...

But remotely!





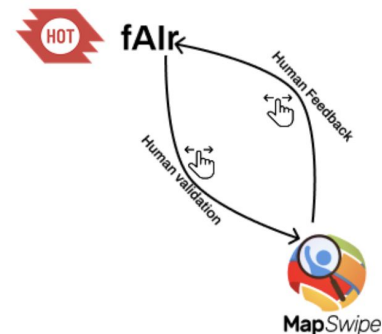
Q2. Should MapSwipe
help validate AI
generated building
footprints?



Conflation Project + fAIr = fAIrSwipe!

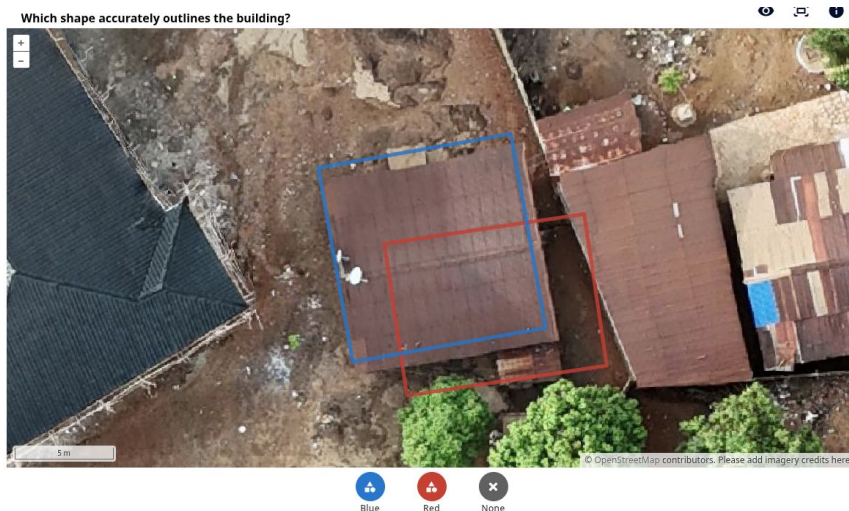
Use the conflation project to compare building footprints from OSM and **fAIr**: HOTOSM's AI modeling tool

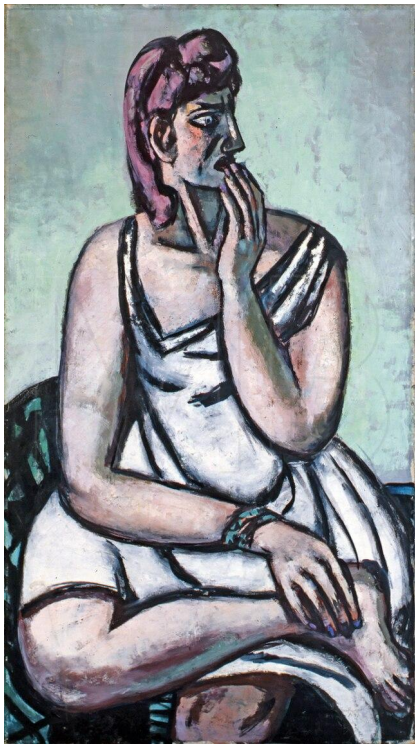
Model showcasing back and forth between fAIr and MapSwipe



In this project type:

- Volunteers decide if the fAIr model or OSM data is better
- OR if neither are well mapped
- Volunteers will **not know** **which** is OSM data and which is the fAIr model
- **Feedback is fed directly to fAIr**





Q3. Combination: MapSwipe/fAlrSwipe project to also push AI generated data to OSM

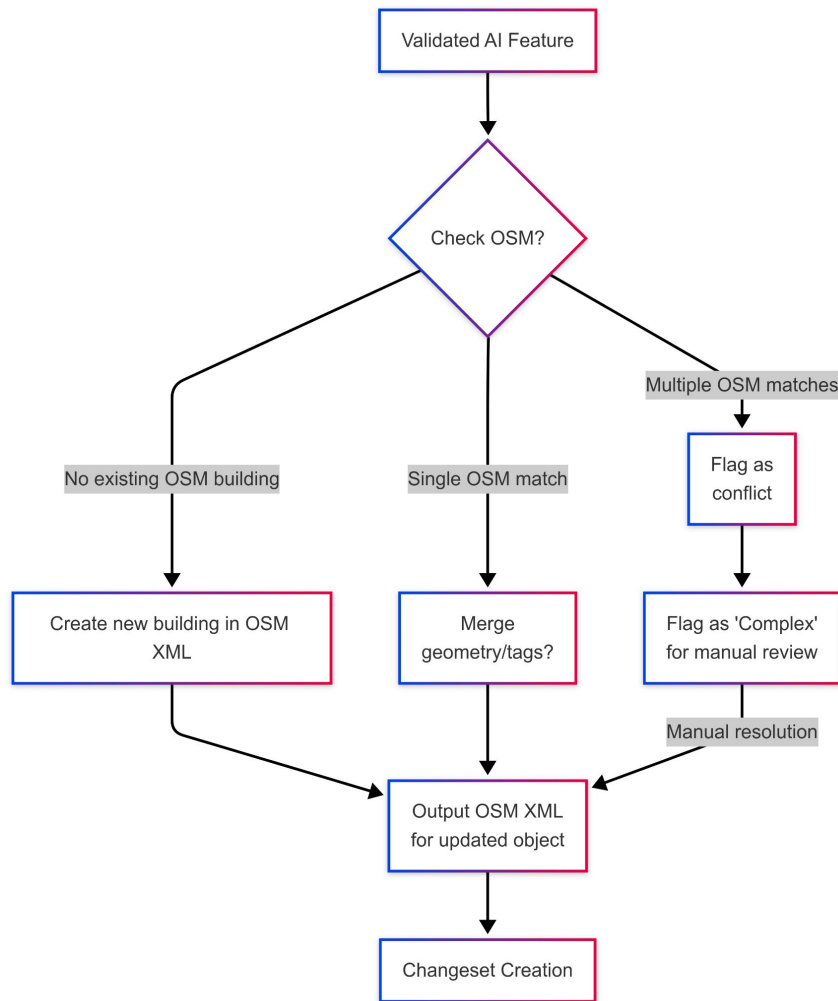


fAIRSwipe AI/ML + Human in the loop validation => OSM

MapSwipe volunteers help decide:

- Whether a building predicted by fAIR is correct, and
- If there is an existing feature in OSM, if it should be updated by the fAIR geometries and / or tags.
- Each feature is **assessed by at least 3 MapSwipers**

The actual “how’s” still to be determined...



Algorithms
written by non
mappers?

Learning
model can't
know all
parts of
world

Who
uses the
data we
create?

AI is bad for
the
environment

Why is AI/ML mapping so controversial?

We train an
algorithm that gets
sold and then the
rug is pulled out
under mappers?

AI = bad?

Intellectual
property
stolen by
companies?

Inexperienced
mappers just
accept AI
suggestions?

Mappers
will be
eventually
replaced?

Data
quality
problems

Work for
beginners
is gone!
How can
we train
and retain
mappers?



MapSwipe Now

No direct connection to OSM

Validation performed manually by volunteers and optionally added to OSM through a separate process

MapSwipe Potential

Direct addition of attribute data to **existing** OSM features (View Streets Project Extension)

Validation performed by AI and **confirmed by humans** before being committed to OSM as an extension of the process (Conflation Project Extension)

Thank you!

Questions?

Nicole Siggins + Benjamin Herfort

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✉ hi@nicolesiggins.com

Try MapSwipe



Comment on my
OSM Diary post



Questions for you:

MapSwipe adding attributes to existing OSM features = good idea?

If 3+ people validate an attribute, who shows as the OSM editor? (One of the users, or a MapSwipe bot?)

If humans refine AI models when **(if ever)** AND **HOW** would it be ok to add AI generated data to OSM ...

AND how could we make sure to validate the quality of that data?

